

Network Switches—An Absolute Necessity for Networking

This buying guide on network switches, which are also called Ethernet switches, will help SMEs and companies in the SOHO segment to design, configure and build their networks with ease.

Popularly known as a switching hub, a network switch is a computer networking device, which connects devices together by using a form of packet switching. It is one asset that's essential for designing, configuring and building your networks. It is a device that lets you connect more Ethernet-based devices to your network without causing any confusion or chaos. It makes your life easy, as each device connected to the network switch can automatically communicate with the other devices already connected to the switch. So before buying a network switch, read on!

Who needs a network/Ethernet switch?

You might be confused about whether or not you need a network switch. You certainly do not require a network switch if you have just one computer to work on, if your Internet usage is limited, and if your computer is directly attached to your ADSL modem. A network switch may not be needed at homes but it is a must in offices or organisations as networking plays a vital role in building network infrastructure.

You need a network switch to build the network's infrastructure. Switches used in such networks are of two types: wired or wireless. Though a lot of people use wireless switches, wired switches are still in high demand and preferred by many. You definitely need a wired network if your wireless connection is low on bandwidth, especially when you are transferring heavy files, want to back up your Mac, your kids want to play online games or download movies, and your spouse wants to stream music—all at the same time, without your network breaking down. For all this, you need an Ethernet network switch.

Factors to keep in mind while selecting a network switch

When you are considering buying a network switch, there are a few factors to be kept in mind. If you want to scale your network in the near future, go for a larger number of ports. Careful planning prior to the purchase will save you money and time, apart from ensuring that you do not end up buying a switch which doesn't suit your requirements. Here are some important factors that should determine the device you select.

1. Number of users: Consider the number of users that you want your network to support. If you only have four or five devices that you need to connect, then an 8-port switch should be enough for your needs. This is how you also end up saving the money and space.

- 2. Basic network infrastructure:** For a small network of up to 50 users, one switch should be enough; whereas, if you want your switch to support more users, you might have to go for multiple switches.
- 3. Determine the role of the switch:** If you plan to build a large network, you should have one or more switches acting as a 'core'. These switches should be fast and able to handle the traffic load. Generally, a Gigabit switch works well as a core switch and access switches (where individual users connect) are likely to be slower than a core switch. If you require to connect a few computers (four or five), a single access switch is what you need.
- 4. Network requirements:** You need to determine your network requirements—do your users need a fast network but with a low latency, or do they require to transfer larger volumes of data? In the latter case, a switch supporting Gigabit Ethernet might be appropriate. Whereas, if the network is used more for Internet and network resource access then a 100 megabit port should be sufficient for your requirements. Figure out how many edge Power over Ethernet (PoE) and PoE+ ports you require and at what speed. Besides, the number of both edge/primary and uplink types and ports are important factors.
- 5. Choosing a vendor:** You may not buy a network switch directly from the manufacturer and may prefer one brand over another. There are quite a few companies offering network switches with different specifications, some of whom are mentioned later in the article. Figuring this out requires some research before you get to know which brand best suits your requirements. Make sure that the company is giving you all the possible support like a 24×7 helpline, hardware replacement, repairs, etc.
- 6. Look for different features:** Most switches are not restricted to having just two or three features, but offer many, which can be considered once your usage and requirements are decided upon. The different options include wired or wireless switches, managed or unmanaged devices, 3-layer capability, etc. These features need to be considered if the switch is being bought for offices, organisations or larger enterprise networks.
- 7. Price vs features:** For a lot of people, price is more important than the features a product offers. But sometimes, considering features over price makes more sense. There are many brands in the market which offer